climate cleanup

Biobased Construction

Certification Protocol for the measurement of net carbon removal benefit

A Climate Cleanup initiative, with support from

Good Energies by Porticus

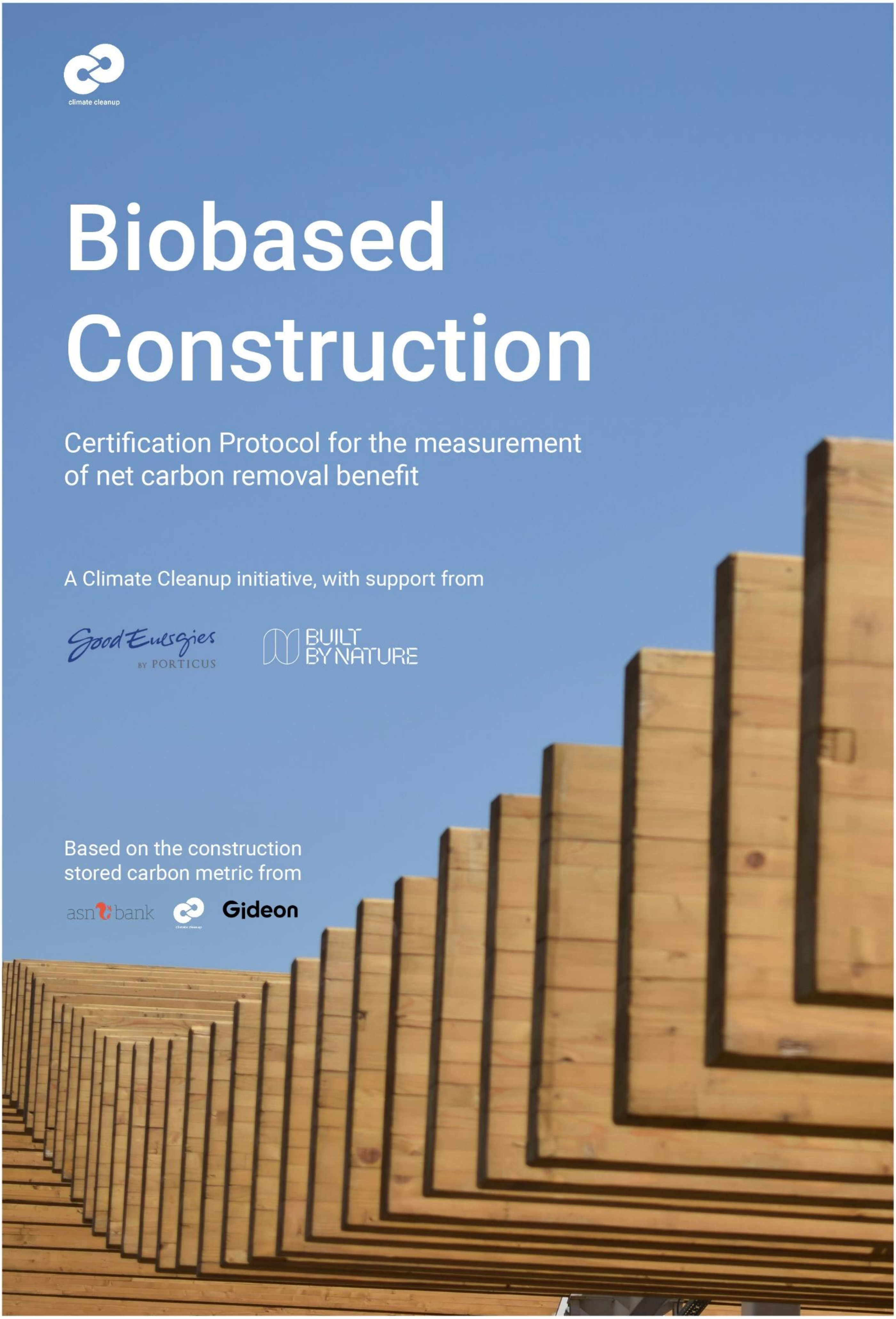
Built by Nature

Based on the construction stored carbon metric from

asn bank

climate cleanup

Gideon



About this document

This Certification Protocol is a result of a research project initiated and stewarded by Climate Cleanup Foundation as a part of its Construction Stored Carbon intervention programme. The development of this Certification Protocol was funded by Built by Nature and Good Energies Foundation as a part of their Construction-stored carbon monetisation initiative. This Certification Protocol is aligned with the Provisional agreement on a Union certification framework for carbon removals between the European Commission and the European Parliament. It also follows the Open Natural Carbon Removal Accounting Guidelines, For more information on the development of this Protocol, visit www.constructionstoredcarbon.org.

This Certification Protocol describes how to robustly certify the Net Carbon Removal Benefit of *Biobased Construction*. Biobased Construction pertains to the use of biobased products in one or more constructions, aimed at or facilitating the long-term storage of biogenic carbon. Given this definition, this Protocol is suitable to certify building-level projects. For a similar protocol on the product manufacturer level, view the Dutch National Carbon Market Foundation's certification method for long-lasting carbon storage in biobased construction materials (in Dutch).

This document consists of three main chapters: (1) general requirements for eligible projects; (2) QUALITY criteria, describing the main criteria to determine the Net Carbon Removal Benefit of eligible projects; (3) MRV requirements, describing the demands for measurement, reporting and verification of claims made and data delivered by operators of Biobased Construction projects.

Authors

Climate Cleanup is a non-profit foundation and social enterprise funded by members and well-aligned partners. Our mission is to reverse climate change by removing 1500 GT CO_2 from the atmosphere with nature-based solutions. We do so by fostering systemic conditions for these nature-based solutions and a regenerative economy. We are grateful to our members; without their support this work cannot be done. We express our gratitude to Built by Nature and Good Energies Foundation for supporting the research project for this Certification Protocol.

Version History

September 2023	First internal draft
February 2024	Second internal draft
31 March 2024	Publication date v1.0
XX April 2024	Start Public Consultation Round 1

Executive Summary

This certification protocol is designed to facilitate the robust quantification and appreciation of the Net Carbon Removal Benefit of a construction project. This benefit is caused by storing carbon for a long time in the constructed building. Carbon is stored in buildings through the use of biomass harvested from plants and trees that have absorbed ('removed') CO_2 from the atmosphere. By constructing buildings using plant-based (or 'biobased') products, the CO_2[1] remains stored. This CO_2 no longer contributes to global warming. Biobased construction thus becomes a climate solution. This protocol explains how to quantify and certify the amount of CO_2 that is stored by using of biobased products in construction.

The carbon stored in eligible biobased products, in CO_2 equivalent, minus a standard baseline, equals the Net Carbon Removal Benefit of a project. The baseline is the amount of carbon that would have been stored anyway in a conventional construction project. This means that every tonne of CO_2 certified as Net Carbon Removal Benefit would not have been removed from the atmosphere and stored in the built environment, if it wasn't for the certified project.

This Protocol is deliberately designed to enable *transition finance* to shift from the use fossil-based products (e.g., concrete, steel), to the use of biobased products. This changes the construction sector from a major polluter into a systemic part of the climate solution. The Protocol can be applied in all phases of construction: in design, under construction and as built, and facilitates products with a first-use lifespan of at least 35 years. If justified properly, a lifespan of 100+ years can be attributed to products certified following this protocol.

A simple process for the certification of a construction project follows these six steps:

1. Make a list of all biobased products in your project that have a first-use lifespan of at least 35 years. The list must include the biobased products' names and their quantities (in kg, m^2 or m^3).
2. Input the products and their quantities into the Excel quantification tool for Net Carbon Removal of your construction project. The tool matches the products to publicly available, verified product data (such as Environmental Product Declarations, EPDs). If verified product data for some of the products you use is not available, contact Climate Cleanup to add it to the tool.
3. Fill in a submission form or project plan for your project. To this form or plan, you add descriptions and details of your project, and you add substantiation and evidence, such as a calculation of the Environmental Performance of your building (Dutch: MPG). You

[1] CO_2 is stored in the form of carbon. Each quantity of carbon in a biobased material can be converted back to a quantity of CO_2 that has been removed from the air. For every 44 grams of CO_2, there are 12 grams of carbon. So, every 12 grams of carbon corresponds to 44 grams of CO_2 removed from the air. The remaining O_2 is oxygen that the tree or plant has exhaled.

can also justify the expected total lifespan of the biobased products used in your building (beyond 35 years, up to 100+ years).

4. Submit both the filled in calculation tool and the submission form to a certifier (such as Oncra). They will validate your certification request and make sure your data is verified once your project is completed and the construction is delivered.
5. After positive validation of your project submission, you sign a contract and receive a preliminary certificate for your building. You can use this for communication purposes and pre-purchase agreements.
6. After positive verification of your project delivery, you receive a full certificate for your building. You can use this to substantiate carbon removal claims and fully realise financial value for your carbon storage.

Contributors

This protocol builds on the work of many. Jeroen Loots at ASN Bank provided its main philosophy already in the development process of the CSC Metric in 2021 – the precursor to this Certification Protocol: keep it Simple, Open and Symbiotic.

The core internal team for this Certification Protocol consisted of Anastasia Vandoorne-Feys, Hanny van Hout, Hajna Tijssen, Tijn Tjoelker, Kevin Priolo, Bart van Valenberg, Sven Jense and Sacha Brons (lead author). Any feedback may be directed to csc@climatecleanup.org or directly to the lead author: sacha@climatecleanup.org.

Other contributions were made by, in no special order, Eelco Leemans, Linda Vosjan, Felix Bruns, Dennis Kerkhoven, Jeroen Loots, Jan-Willem van der Groep, Norbert Schotte, Lars Hillewaere, Pablo van der Lugt, Erik Gathier, Maarten Markus, Malika van de Weerd, Hala Abi Saad, Willem van Genugten, Alma Krug, Pieter van den Wittenboer, Jurrian Knijtijzer, Sam Knetsch, Reno Mol, Jannes Nelissen, Peter Fraanje, Julien Vincelot, Karianne Kraaijestein, Onno Dwars, Rosa Bos, Jeroen Pels, Matija Kajić, Simon Scholl, Sander Teeuwen, Samy Porteron, Ruben Lentz, Mark Compeer, Robert Winkel, Robert Platje, Radboud van Hemel, Sanne Vogelzang, Thijs de Goede, Jip van Grinsven, Ralf van Tongeren, Jim Teunizen, Mike van Vliet, Sybren Bosch, Laetitia Nossek, Ruben Zonnevijlle, Frank Vaset, Wolf Bierens, Gert-Jan Nabuurs, Sven van Baren, Gerben Sinke, Patrick de Baat, Ruben Schmitz, Joost Blankendaal, Marco Vermeulen, Hidde Proost, Deen Mondt, Bob van der Zande, Imme Groet, Jurriaan van Stigt, Reinoud van der Zijde, Yaël Ben-Basat, Nina van der Giessen, Gijs Kaper, Sandra Nap, Pepijn Duijvestein, Lidewij Lenders, Eric de Munck, Lucy Gibbons, Wihan Bekker, Bob Jansen, Maurice Breg, Ward Rennen, Arjan Alkema, and others.

We thank each and every contributor for their feedback, insight, critique, compassion and drive to transform the built environment into a natural carbon sink.

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Abbreviations

BSF	Baseline Storage Factor
C	Carbon
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
CR	Carbon Removal
CSC	Construction Stored Carbon
EPD	Environmental Product Declaration
GFA	Gross Floor Area
GHG	Greenhouse Gas
GWP	Global Warming Potential
LCA	Life Cycle Assessment
MPG	Environmental Performance Building (Dutch: MilieuPrestatie Gebouwen)
MRV	Measurement, Reporting, Verification
NCRB	Net Carbon Removal Benefit
RSL	Reference Service Life

Glossary

For the purposes of this Certification Protocol, the following definitions apply:

a. ‘additionality’ means the extent to which the certification of a carbon removal activity leads to an additional benefit to the climate beyond a baseline situation without certification;
b. ‘baseline scenario’ means the standard carbon removal performance of comparable activities in similar social, economic, environmental and technological circumstances and [taking] into account the geographical context;
c. ‘biobased products’ means products intended for long-term use in or as construction elements, consisting of at least 70% renewable, continually replenishable biomass determined according to EN16575:2014*;*
d. ‘biogenic carbon’ means carbon dioxide removed from the atmosphere through photosynthesis and stored as carbon in biomass;
e. ‘carbon removal’ means either the storage of atmospheric or biogenic carbon within geological carbon pools, biogenic carbon pools, long-lasting products and materials, and the marine environment, or the reduction of carbon release from a biogenic carbon pool to the atmosphere;
f. ‘carbon removal activity’ means one or more practices or processes carried out by operators resulting in permanent carbon storage, enhancing carbon capture in a biogenic carbon pool, reducing the release of carbon from a biogenic carbon pool to the atmosphere, or storing atmospheric or biogenic carbon in long-lasting products or materials;
g. ‘carbon removal system’ means the sum of all Greenhouse Gas emissions and removals related to the project that are included within the scope of certification;
h. ‘carbon removal unit’ means one tonne of certified net carbon removal benefit generated by a carbon removal activity and registered by a certification scheme;
i. ‘certificate’ means a document published by a certification body or scheme confirming the Net Carbon Removal Benefit of a project operated by an operator;
j. ‘certification body’ means an independent, accredited or recognised conformity assessment body that has concluded an agreement with a certification scheme to carry out certification audits and issue certificates;
k. ‘certification scheme’ means a scheme managed by a private or public organisation that oversees the certification of operators or group of operators;
l. ‘construction stored carbon’ is a metric for the physical storage of biogenic carbon in buildings or structures expressed in tonnes of CO_2;

m. 'in design' means the first stage of Biobased Construction projects in which only a conceptual design with a provisional bill of materials is available;
n. 'under construction' means the second stage of Biobased Construction projects in which a building permit and definitive bill of materials is available, but the construction project has not been completed;
o. 'as built' means the third and final stage of Biobased Construction projects in which the project has been completed and delivered, and is in the process of verification or has been verified;
p. 'long-term storage' means justifiably expected storage of carbon dioxide removed from the atmosphere for at least thirty-five (35) years[2], and potentially more than one hundred (100) years[3];
q. 'monitoring' means performing checks at a regular interval to determine whether the quantified and/or delivered Net Carbon Removal Benefit of a project is still valid in reality, with special regard to whether the quantified biogenic carbon is still stored;
r. 'Net Carbon Removal Benefit' means the effect of a carbon removal activity on the concentration of atmospheric greenhouse gases compared to a reference ('business-as-usual') scenario, excluding any avoided emissions or substitution effects;
s. 'operator' means any legal or physical person who operates or controls a carbon removal activity, or to whom decisive economic power over the technical functioning of the activity has been delegated;
t. 'group of operators' means a legal entity that represents more than one operator and is responsible for ensuring that those operators comply with this certification protocol;
u. 'risk of reversal' means the risk that any delivered carbon removals are emitted into the atmosphere before the minimal total lifespan of 100 years;
v. 'scope of certification' means the boundaries in space and time within which information related to the project is considered under this Certification Protocol;
w. 'upfront carbon' means the carbon dioxide emitted in the production and construction phase of products and materials used in a construction process, otherwise known as embodied carbon in modules A1-A5 of construction products and materials, excluding negative biogenic emissions (removals);
x. 'validation' means the process of screening and evaluating a project submission and the data delivered by operators of that project;
y. 'verification' means the process in which an independent third party with relevant expertise verifies whether the promised Net Carbon Removal Benefit is delivered.

[2] As determined by the European Commission and European Parliament in their Provisional agreement on a Union certification framework for carbon removals.
[3] Commonly referred to as 'permanent removal' (e.g., see IPCC. (2000). Land Use, Land-Use Change and Forestry: 2.3.6.3. Equivalence Time and Ton-Years).

1. General Activity Requirements

1.1 Eligible Projects

Projects eligible for certification using this protocol shall meet the following criteria:

a. The carbon removal activity of the project is defined as the use of sustainably sourced **biobased products** in the construction, renovation or alteration of buildings or structures, aimed at or facilitating the **long-term storage** of biogenic carbon. As such, the scope of certification is limited to biobased construction products that have a minimal lifespan in first use of 35 years.

b. The project meets the minimum additionality and sustainability criteria outlined in paragraphs 2.2 and 2.4.

c. The construction project facilitating the carbon removal either:
   i. Has a completed (temporary) design but no official building permit, in which case the project is labelled **'in design'**. Projects 'in design' do not generate any tradable carbon removal units. Operators of projects 'in design' may request a *Validated Calculation Statement*, which can be used, for example, as justification of potential carbon removal claims made in tender applications.
   ii. Has a definitive design that is approved by a commissioning party (i.e., through a building permit), and has an expected date of completion within five years of the date of submission. In this case, the project is labelled **'under construction'**. Operators of projects 'under construction' may pre-sell up to fifty percent (50%) of their quantified net carbon removal benefit.
   iii. Has been completed up to one year before the date of submission and has a realised Net Carbon Removal Benefit. In this case, the project is labelled **'as built'**. Operators of projects 'as built' may sell one-hundred percent (100%) of the generated carbon removal units.

i) in design ii) under construction iii) as built

Figure 1. Visualisation of construction stages eligible for validation or (provisional) certification.

Table 1. Characteristics of construction stages eligible for certification under this protocol.

Characteristic	(a) in design	(b) under construction	(c) as built
Fits the definition of Biobased Construction as a Carbon Removal activity	✓	✓	✓
Has a design with a bill of materials	✓	✓	✓
May request a Validated Calculation Statement	✓	✓	✓
Has a definitive design and a building permit	✗	✓	✓
Is expected to be completed within five years of submitting a certification request	✗	✓	✓
May gain provisional certification and claim or pre-sell 50% of the generated carbon removal units	✗	✓	✓
Has been completed up to one year before submission and has delivered a Net Carbon Removal Benefit	✗	✗	✓
May gain full certification and claim or sell 100% of the generated carbon removal units	✗	✗	✓
May gain full certification even if the building has been completed over one year before submission	✗	✗	✗

1.2 Eligible Project Operators

To be eligible for receiving certification for one or more of their 'Biobased Construction' projects, operators must meet the following criteria:

a. At least one of the operators of the project must have legal ownership and/or control over the design, contracting, execution, delivery, and ownership or sale of the construction in which the biobased products are used. Legal ownership and/or control over these different construction stages may be divided between operators.

b. In case of a grouped project in which one operator, acting as the registration preparer and/or certificate beneficiary, represents (multiple) other operators, written agreements on liability and certificate ownership must be signed between all operators.

1.3 System boundaries and scope of certification

The **carbon removal system** is defined as the sum of all Greenhouse Gas (GHG) emissions and removals related to the project that are included within the **scope of certification**. Figure (2) shows the carbon removal system boundaries for 'Biobased Construction'. The following criteria apply to the scope of certification:

a. The carbon removal system is limited to modules A1-A5 of a standard Life Cycle Assessment (LCA).
b. The only removals included in the system boundaries are negative biogenic emissions in module A1 of the LCAs of eligible biobased products used in the project.
c. The emissions included in the system boundaries are all emissions in modules A1-A5 of the LCA of the eligible biobased products used in the project.
d. Any GHG emissions or removals beyond these boundaries (i.e. after module A5) lie outside of the scope of certification and therefore do not impact the quantification of Net Carbon Removal Benefit for 'Biobased Construction' projects.

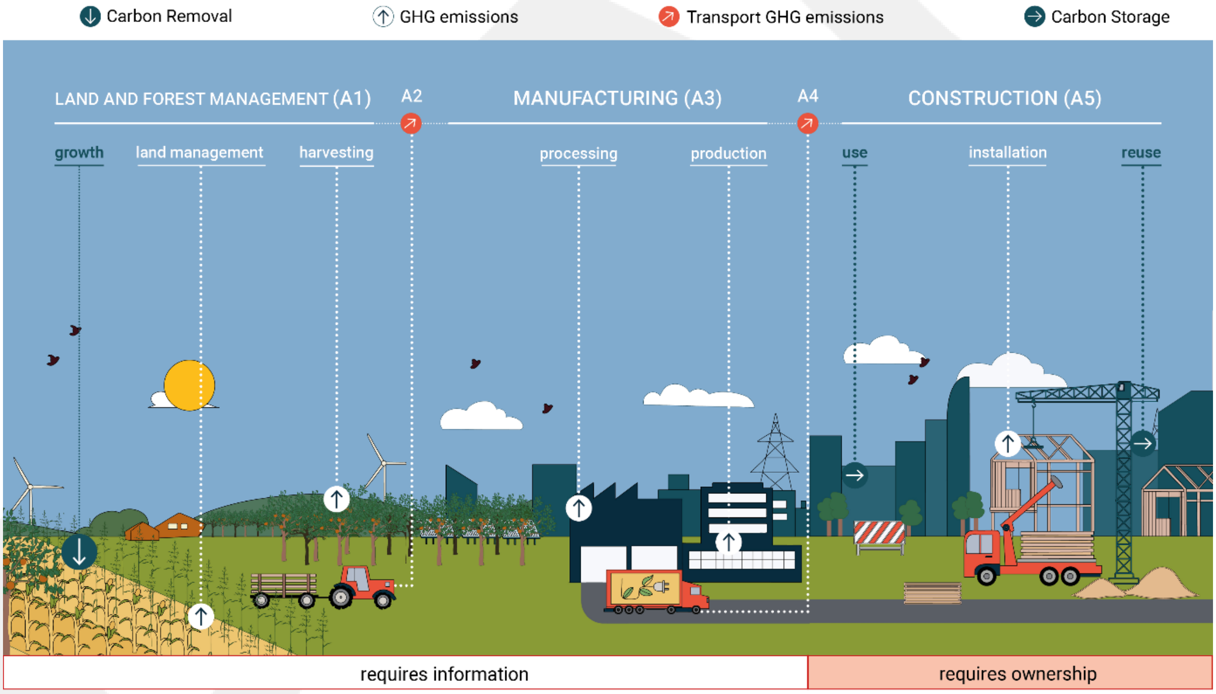

Figure 2. Carbon removal system of 'Biobased Construction', including the removal, storage and emissions of CO_2-equivalents in modules A1 to A5 of the life cycle of construction products.

2. 'QUALITY' Criteria

2.1 Quantification

Quantification of an eligible project's **Net Carbon Removal Benefit** shall be performed according to the criteria outlined in paragraphs 2.1.1-2.1.4. Paragraph 2.1.1. describes the general formula for an eligible project's Net Carbon Removal Benefit, Formula (1). Paragraphs 2.1.2-2.1.4 describe the methods, formulas and criteria for quantification of each of the elements of Formula (1). The general criteria for the use of Formula (1) and the interpretation of its elements are derived from the Provisional agreement on a Union certification framework for carbon removals between the European Commission and the European Parliament.

2.1.1 Net Carbon Removal Benefit

a. Operators shall use Formula (1) to quantify the Net Carbon Removal Benefit of their project.

(1) $$NCRB = CR_{baseline} - CR_{total} - GHG_{increase} > 0$$

symbol	description	unit
NCRB	Net carbon removal benefit	t CO_2e
CR_baseline	Carbon removals under the baseline scenario	(-) t CO_2e
CR_total	Total Project carbon removals	(-) t CO_2e
GHG_increase	Direct and indirect increase in GHG emissions	(+) t CO_2e

b. In using Formula (1), operators shall comply with the following criteria:

   i. Quantities referred to in Formula (1) shall be shall be designated with a negative sign (-) if they are net greenhouse gas removals (i.e. CR_baseline and CR_total) and with a positive sign (+) if they are net greenhouse gas emissions (i.e. GHG_increase);

   ii. Quantities referred to in Formula (1) shall be expressed in tonnes of carbon dioxide equivalent (t CO_2e).

   iii. Operators shall highlight any cases of uncertainty regarding the quantification of any of the elements of Formula (1) and shall make justifiably conservative estimates when exact quantities are not available. For example, when a range of values is available for the quantification of an element or sub-element of Formula (1), operators shall choose a value on the lower end of that range.

2.1.2 Carbon removals under the baseline scenario

The **baseline scenario** for Biobased Construction is defined as the standard carbon removal performance of construction activities in similar social, economic, environmental and technological circumstances as the project, taking into account the geographical context. Carbon removal and storage in conventional constructions depends on the region and typology of the construction. Baseline scenarios therefore have to be constructed per region and per typology.

Scientific or other independent data on standard carbon storage in conventional constructions per region and per typology are not yet available. After testing and reviewing several options, the following approach for quantifying the baseline scenario is currently deemed best. Until more robust data is available, conservative estimates shall be made based on available data.

a. To determine the baseline scenario operators may choose one of three approaches:
   i. Operators shall use a predetermined standard baseline storage factor for their project's region if shown in Table 2.
   ii. If there is no predetermined baseline storage factor for their project's region, operators shall quantify their own baseline storage factor for their project, by finding or developing reference scenarios that reflect a conventional materialisation of their project. These reference scenarios shall be subject to peer-review and verification.
   iii. If a predetermined baseline storage factor is available for their project, but operators deem this value inaccurate for their project's context, they may also choose approach (ii) if satisfactory justification is provided.

b. Table 2 shows standard baseline storage factors per typology for currently available regions, to be used in approach 2.1.2a(i). These baseline storage factors were calculated using the CO_2-density of biobased products that have the potential to last 35 years and are commonly used in conventional constructions. In order of contribution to carbon storage these are: slanted roofs, wooden staircases, wooden window frames, battens and doors. The CO_2 density of these elements was multiplied by their occurrence in reference projects with known GFA[4]. Most CO_2 in conventional constructions is stored in slanted roofs. For typologies without a slanted roof, carbon storage is minimal. A conservative estimate of 10 kg CO_2e / m^2 GFA was chosen for these other typologies.

Table 2. Predetermined baseline storage factors for available regions.

Region	Unit for carbon storage	Residential: slanted roof	Residential: flat roof	Residential: multi-storey	Utility: office	Utility: other
The Netherlands	kg CO_2e / m^2 GFA	32	10	10	10	10

[4] These reference projects are protected under NDA. This highlights the need for open-source, independently generated data on carbon storage in conventional constructions.

c. To quantify their own baseline storage factor following approaches 2.1.2a(ii) and (iii), operators shall use Formula (2).

$$(2)\quad BSF = \frac{Biobased_{ref}}{GFA_{ref}} \times CF \times \frac{44}{12}$$

symbol	description	unit
BSF	Baseline Storage Factor in a specific region	kg CO_2e / m^2
Biobased_ref*	Volume of biobased products used in the reference project(s)	m^3
GFA_ref	Gross floor area of the reference project(s)	m^2
CF	Conversion factor from m^3 to carbon[5]	Mg C / m^3
44/12	Carbon to CO_2 conversion factor	kg CO_2e / kg C

**Must be considered at appropriate moisture levels as per EN 16449*

d. To quantify carbon removals under the baseline scenario, operators shall use Formula (2) in combination with their chosen baseline storage factor.

$$(3)\qquad CR_{baseline} = \frac{-1}{1000} \times BSF \times GFA_{project}$$

symbol	description	unit
CR_baseline	Carbon removals under the baseline scenario	(-) t CO_2e
BSF	Baseline Storage Factor for the project	kg CO_2e / m^2 GFA
GFA_project	Gross Floor Area of the project	m^2

[5] A standard carbon conversion factor of 0.229 can be used if the reference project(s) mostly use wood. This is based on the C conversion factor (per air dry volume) of sawn wood (aggregate) as per "table 2.8.1: default conversion factors for the default HWP categories and their subcategories." in IPCC. (2014). 2013 Revised Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. in T.

2.1.3 Total project carbon removals

a. Total project carbon removals for a Biobased Construction project are defined as the sum of the Construction Stored Carbon (CSC) values of all eligible biobased products used in the project, expressed in tonnes of CO_2-equivalent[6]. Depending on the available product-specific data, the following variants of Formula (4) shall be used to quantify the CSC value of each eligible biobased product used, in order of priority:

i. The product has an EPD in EN 15804+A2 format and Biogenic Carbon content in **$kgCO_2e$**/functional unit is given by the product's EPD. In this case;

(4.i) $$CSC = CO2_{biogenic} \times N$$

symbol	description	unit
CSC	Construction Stored Carbon	kg CO_2e / 100 years
CO2_biogenic	Biogenic Carbon Content	kg CO_2e / unit
N	Quantity of product used in the project	kg, m^3, m^2 or m

ii. The product has an EPD in EN 15804+A2 format and Biogenic Carbon content in **kgC**/functional unit is given by the product's EPD. In this case;

(4.ii) $$CSC = C_{biogenic} \times N \times \frac{44}{12}$$

symbol	description	unit
CSC	Construction Stored Carbon	kg CO_2e / 100 years
C_biogenic	Biogenic Carbon Content	kg C / unit
N	Quantity of product used in the project	kg, m^3, m^2 or m
44/12	Carbon to CO_2 conversion factor	kg CO_2e / kg C

iii. No specific value is given for Biogenic Carbon content and the product's EPD is in EN 15804+A2 format. The Biogenic Carbon content should be estimated based on stages A1 to A3 from GWP - biogenic. In this case;

(4.iii) $$CSC = -1 \times (GWP_{A1}biogenic + GWP_{A2}biogenic + GWP_{A3}biogenic) \times N$$

symbol	description	unit
CSC	Construction Stored Carbon	kg CO_2e / 100 years
GWP_{A1}biogenic	Negative biogenic emissions in raw materials supply	kg CO_2e / unit
GWP_{A2}biogenic	Positive biogenic emissions in transport to factory	kg CO_2e / unit
GWP_{A3}biogenic	Positive biogenic emissions in product manufacturing	kg CO_2e / unit
N	Quantity of product used in the project	kg, m^3, m^2 or m

[6] For clarification of the Construction Stored Carbon metric, view https://climatecleanup.org/constructionstoredcarbon/metric/ and this report by SGS Search for the Dutch Ministry of the Interior and Kingdom Relations. The lifespan multiplication factor in these metric has been disregarded in this

iv. No specific value is given for Biogenic Carbon content, the product's EPD is <u>not</u> in EN 15804+A2 format and the product is timber-based (i.e. at least 95% woody biomass). In this case, Biogenic Carbon content may be estimated to be 0.45 kg C / kg product as per EN 16449. In this case;

(4.iv)
$$CSC = V \times \rho \times 0.45 \times N \times \frac{44}{12}$$

symbol	description	unit
CSC	Construction Stored Carbon	kg CO_2e / 100 years
V*	Volume of timber-based product	m^3
ρ*	Density of timber-based product	kg/m^3
0.45	Simplified carbon content of the timber-based product	kg C / kg product
N	Quantity of product used in the project	kg, m^3, m^2 or m
44 / 12	Carbon to CO_2 conversion factor	kg CO_2e / kg C

**Must be considered at appropriate moisture levels as per EN 16449*

v. No specific value is given for Biogenic Carbon content and the product's EPD is <u>not</u> in EN 15804+A2 format and the product is <u>not</u> timber-based (i.e. less than 95% woody biomass). In this case, the same method applies as in formula (4.iv), but $\frac{m_c}{m_{tot}}$ must be determined through an independent product-specific study. In this case;

(4.v)
$$CSC = V \times \rho \times \frac{m_c}{m_{tot}} \times N \times \frac{44}{12}$$

symbol	description	unit
CSC	Construction Stored Carbon	kg CO_2e / 100 years
V*	Volume of timber-based product	m^3
ρ*	Density of timber-based product	kg/m^3
m_c / m_tot	Simplified carbon content of the timber-based product	kg C / kg product
N	Quantity of product used in the project	kg, m^3, m^2 or m
44 / 12	Carbon to CO_2 conversion factor	kg CO_2e / kg C

**Must be considered at the same moisture levels as per the quantification of* $\frac{m_c}{m_{tot}}$

Regardless of the variant of Formula (4) chosen, Formula (5) shall be used to quantify total project carbon removals as a sum of the CSC-values of each eligible biobased product used.

(5)
$$CR_{total} = \frac{1}{1000} \times \sum_{i=1}^{N} CSC_i$$

symbol	description	unit
CR_total	Total project carbon removals	(-) t CO_2e
CSC_i	Construction Stored Carbon of biobased product i	kg CO_2e
N	Total number of biobased products used	kg, m^3, m^2 or m

2.1.4 Direct and indirect increase in GHG emissions

Compared to the baseline scenario, any direct or indirect increase in GHG emissions (GHG_increase) caused by the project must be subtracted from the project's Net Carbon Removal Benefit, as shown in Formula (1).

Table 3 shows the GHG emission sources included in the scope of this Certification Protocol as defined in paragraph 1.3.

a. The baseline scenario for Biobased Construction, as described in paragraph 2.1.2, is predominantly shaped by the use of conventional construction products. Within the scope as shown in Table 3, biobased products generally have less embodied emissions and therefore do not contribute to an increase in GHG emissions. Additionally, no indirect increase in GHG emissions (leakage) occurs, under the condition that biomass for the eligible biobased products used is sustainably sourced. Instead, these products avoid emissions by substituting conventional materials[7]. Avoided emissions are not included in the scope of this Certification Protocol. Therefore, given the scope and baseline scenario mentioned, GHG_increase for Biobased Construction is defined in Formula (6).

$$(6) \qquad GHG_{increase} = 0$$

Table 3. GHG sources included in the calculation of UC_project and UC_baseline

Emission Source	Included?	Explanation
A1-3 GWP-fossil	Yes	Always included.
A1-3 GWP-biogenic	No	Included in quantification of CR_total
A1-3 GWP-luluc	Yes	Always included.
A4-5 GWP-fossil, GWP-biogenic & GWP-luluc	Yes	Always included.
B1, B6 & B7 GWP-fossil, GWP-biogenic & GWP-luluc	No	Not included, covered by sustainability criteria 2.4.1a and 2.4.1b.
B2-B5 GWP-fossil, GWP-biogenic & GWP-luluc	No	Not included, irrelevant for eligible products and beyond the scope of this certification.
C1-4 GWP-fossil, GWP-biogenic & GWP-luluc	No	Not included, beyond the scope of this certification.
Module D	No	Not included, module D is only potentially used as substantiation of reusability or lifespan claims.

[7] For scientific literature on avoided emissions, e.g., see Arehart *et al.*, (2021) or Leskinen *et al.*, (2018).

2.2 Additionality

Biobased Construction projects are deemed additional and therefore eligible for certification if operators comply with the following additionality criteria:

a. Any carbon removal and/or storage **directly** subsidised or required by law, either in total tonnes of CO_2e or as a percentage of total project carbon removals (CR_total), shall be subtracted from the activity's Net Carbon Removal Benefit.

b. For the purposes of criterion 2.2a, a subsidy measure is considered **directly** relevant to carbon removal when it specifically target tonnes of CO_2e removed or similar, or when the subsidy's demands are **only** achievable through the use of biobased products or the removal and storage of carbon. Any other subsidy or statutory requirement is deemed irrelevant for this certification protocol. The Dutch MIA\Vamill subsidy is deemed irrelevant for this certification protocol as of this version of the Certification Protocol.

c. The carbon removals associated with the project shall not be double counted (i.e. no two certificates are issued for the same tonne of CO_2e removed). Operators shall be held liable for any occurrence of double counting. Three types of double counting shall be considered:

   i. Operators shall not certify the Net Carbon Removal Benefit of their product under any other carbon removal certification scheme.

   ii. Operators shall ensure to the best of their abilities that the manufacturers of the eligible biobased products used in the project have not yet claimed and/or certified the Net Carbon Removal Benefit generated by the products. Any products of which the Net Carbon Removal Benefit has already been claimed will be excluded from certification following this protocol.

   iii. All carbon removal units generated with the project and sold to a third party shall not be used by operators themselves for climate neutrality claims, unless the relevant purchase agreement(s) explicitly state(s) that the buyer refrains from these claims.

2.3 Long-term storage

To ensure the **long-term storage** of carbon, operators must meet the following criteria:

a. The minimum first lifespan of each eligible biobased product used in the project shall equal 35 years, in accordance with the [Provisional agreement on a Union certification framework for carbon removals]() by the European Commission and Parliament.

b. The minimum first lifespan of each eligible biobased product used in the project must have been determined through independent assessment. The Reference Service Life (RSL) of a product as declared in an Environmental Product Declaration (EPD) is an eligible data source.

c. For each eligible biobased product used in the project with a lower first lifespan than the building's lifespan, operators shall describe and justify a maintenance and end-of-life scenario.

d. Operators may substantiate a total lifespan of their biobased products used, by including a reused lifespan. The total lifespan of a biobased product may be quantified using Formula (7):

$$L_{total} = L_{use} + v \times L_{reuse} \quad (7)$$

Symbol	description	unit
L_total	Total product lifespan	years
L_use	Lifespan of product in first application	years
v	Correction variable for product disassembly potential	-
L_reuse	Lifespan of product in reused applications	years

e. In the quantification of total product lifespan (L_total) for a biobased product, the elements of Formula (7) shall be substantiated as follows:
    i. The product's lifespan in first application (L_use) shall be determined using the product's Reference Service Life (RSL) as declared in a valid EPD, independent technical report, or similar.
    ii. The correction variable for product disassembly potential (v) shall equal zero (0) when no data is provided as substantiation and shall range between zero (0) and one (1) depending on the disassembly potential of the product or its building layer. This reusability shall be substantiated by calculating the Disassembly

Potential of each eligible biobased product used, using the 'Circular Buildings' Disassembly Potential Measurement Method[8] or similar.

iii. The product's lifespan in reused applications (L_reuse) shall be determined using the product's technical lifespan as independently shown in scientific literature, technical reports or similar. Operators may also use take-back guarantees from suppliers as substantiation of L_reuse. **The maximum value for L_reuse shall equal the product's lifespan in first application (L_use).**

f. After operators deliver the project and its Net Carbon Removal Benefit, the risk of reversal of any carbon stored must be monitored. Operators shall choose **one** of the following two options for monitoring the risk of reversal:

   i. Monitoring shall be performed by the certification scheme, or an independent third party appointed by the certification scheme. The certification scheme shall periodically publish an audit report that verified the delivered Net Carbon Removal Benefit of the project. Only publicly available data sources, such as remote imagery, cadastral data, confirmed permits, or external inspection, shall be used for audits.

   ii. The building owner commits to monitoring any release of stored biogenic carbon with a monitoring cadence of at most five (5) years. Every five years or less, the building owner shall report on the status of the project's quantified and delivered Net Carbon Removal Benefit. This report shall be verified independently.

g. Monitoring requirements may be nullified if region-specific statutory requirements ensure the long-term storage of biogenic carbon in the project for at least the duration of the monitoring period. For example, if statutory law requires the 100% reuse of biobased construction products, the monitoring requirements become obsolete.

[8] Alba Concepts, Dutch Green Building Council, Netherlands Enterprise Agency, & W/E Adviseurs (2021). *Circular Buildings. Disassembly Potential Measurement Method*. For background information (Dutch and English), see here. For a direct download link (English version), click here.

2.4 Sustainability

2.4.1 Minimum requirements

a. To comply with the minimum sustainability requirements of this Certification Protocol, the project shall have at least a neutral impact on the following six sustainability objectives:
   i. Climate change mitigation;
   ii. Climate change adaptation;
   iii. Sustainable use and protection of water and marine resources;
   iv. Transition to a circular economy;
   v. Pollution prevention and control;
   vi. Protection and restoration of biodiversity and ecosystems.

   To justify a neutral impact on these objectives, operators shall comply with criteria 2.4.1b-d.

b. Operators shall provide evidence of compliance with national statutory requirements for building project sustainability (e.g., MPG in the Netherlands, RE2020 in France). If no harmonised or standardised national sustainability requirements exist for the project, operators may alternatively provide an (inter)nationally recognised sustainability certificate of choice (e.g., BREAAM, GPR Gebouw). In this case, the certification scheme shall determine whether the attained sustainability certificate complies with the minimum sustainability requirements.

c. Operators shall provide evidence that the building's energy performance meets national statutory requirements (such as BENG in The Netherlands) or similar.

d. Operators shall provide evidence of at least neutral biodiversity impact of all eligible biobased products used. This may be done in two ways, depending on the type of product used:
   i. For wood-based products, operators shall prove that the product is certified according to FSC, PEFC or similar certification;
   ii. For other biobased products, operators shall prove within reasonable assurance that the production of raw materials for these products has not contributed to deforestation or destruction of vulnerable ecosystems. Raw materials sourced from the European Union are deemed to comply with this demand unless there is reason to suspect otherwise.

2.4.2 Sustainability co-benefits

a. Operators may report on any additional benefits generated by the project beyond the minimum sustainability requirements outlined in paragraph 2.4.1. These co-benefits shall be communicated transparently by certification schemes in a clear and concise manner. The following criteria apply for co-benefit reporting on the six sustainability objectives as listed in criterion 2.4.1a:
   i. For climate change mitigation, operators may report on the expected avoided emissions resulting from their project, using internationally accepted scientific data and estimates on avoided emissions per product category and region;
   ii. For climate change adaptation, operators may report on the interventions taken at the building level that improve climate change adaptation;
   iii. For sustainable use and protection of water and marine resources, operators may describe how the project improves water use and water retention compared to conventional constructions;
   iv. For the transition to a circular economy, operators may show how the project uses circular materials and/or has improved circularity (i.e. through design for disassembly), based on a quantification of Product Disassembly Potential following criteria 2.3d and 2.3e;
   v. For pollution prevention and control, operators may show calculations of the emissions of chemical compounds beyond carbon dioxide equivalents related to the project;
   vi. For the protection and restoration of biodiversity and ecosystems, operators may show how the project improves biodiversity and/or ecosystem integrity at the building level and/or throughout the value chains of the products used in the project. Any biodiversity claims made must be independently verified before submission.

b. Any additional achieved sustainability certification (e.g., the Dutch 'Het Nieuwe Normaal') on the building level may be submitted and shall be shown on certification documents and/or project pages related to the project.

c. Operators may report on any other co-benefits (e.g., social or economic co-benefits) generated by the project beyond the sustainability objective outlined in criterion 2.4.1a. Their description of other co-benefits shall be transparently communicated by the certification scheme.

3. Measurement, Reporting and Verification Requirements

3.1 Measurement

When submitting a project for certification following this Certification Protocol, operators shall comply with the following criteria regarding measurement and data collection:

a. Before requesting a Validated Calculation Statement for their project 'in design', operators shall ensure the following is data is measured and/or collected:
   i. The gross floor area (GFA) of the project to be certified;
   ii. A list of eligible biobased products used in the project, including the quantity of each biobased product used, measured in the respective product's functional unit;
   iii. Whether or not any of the biobased products used have already received Carbon Removal certification or are subject to similar carbon dioxide claims;
   iv. Product-specific data on biogenic carbon storage for each of the biobased products used in the project conforming to the criteria outlined in paragraph 2.1.3;
   v. Building-level quantification of CR_baseline as per the criteria outlined in paragraph 2.1.2;
   vi. Building-level quantification of GHG_increase based on a chosen GHG_baseline and the MPG of the construction project as per the criteria outlined in paragraph 2.1.4;
   vii. Description of any statutorily required or subsidised carbon removals relevant to the project;
   viii. The total lifespan of each biobased product used, quantified according to the criteria outlined in paragraph 2.3;

b. Before requesting a Carbon Removal Certificate for their project 'under construction', operators shall ensure the following data is measured and/or collected in addition to the data requirements outlined in criterion 3.1a:
   i. Proof of building permit;
   ii. Proof of compliance with relevant (inter)national and regional environmental law;
   iii. Final design of the construction project as submitted to obtain the building permit as referred to in point (i) of criterion 3.1b;

iv. Final MPG of the construction project as submitted to obtain the building permit as referred to in point (i) of criterion 3.1b;

v. Expected timeline of the project including at least the starting date of construction work and the expected date of completion of the project;

vi. When transitioning from a project 'in design' to a project 'under construction' and relevant: updated quantification of the project's Net Carbon Removal Benefit with a detailed description of the updated calculations;

vii. When transitioning from a project 'in design' to a project 'under construction' and relevant: updates to any of the information delivered for the 'in design' stage of the project, in case any of the information submitted for the 'in design' stage of the project is no longer accurate.

c. Before requesting a Carbon Removal Certificate for their project 'as built', operators shall ensure the following data is measured and/or collected in addition to the data requirements outlined in criteria 3.1a and 3.1b:

   i. Supplier invoices or similar evidence that the project's quantified Net Carbon Removal Benefit has been delivered in reality;

   ii. Photographic evidence of the project development and construction process, including but not limited to pictures of the start and end of the construction process;

   iii. Date and proof of completion;

   iv. Description and proof of the final ownership of the completed construction;

   v. Quantification of the Project's delivered Net Carbon Removal Benefit based on the product quantities as substantiated following point (i) of criterion 3.1c.

3.2 Reporting

When reporting on the Net Carbon Removal Benefit of the project, operators shall comply with the following criteria:

a. Operators shall submit data gathered in accordance with the criteria outlined in paragraph 3.1 by writing a project plan or filling in a project submission form as provided by the certification scheme. Any proofs shall be separately submitted as appendices.

b. Operators shall re-submit a project form or project plan when their project reaches the next stage in the construction process (i.e. from 'in design' to 'under construction' and from 'under construction to 'as built').
c. During the 'under construction' stage of the project, operators shall monitor and report on any significant changes (i.e. >5%) in the quantified Net Carbon Removal Benefit of the project.
d. During the 'under construction' stage of the project, operators shall monitor and report on any significant changes (i.e. >5%) in the timeline of the project.

3.3 Verification

In addition to validation of project plans or submission forms by the certification scheme and/or body, any project awaiting full certification shall be verified by an independent third party with relevant expertise to ensure any certified Net Carbon Removal Benefit is delivered in reality. To that end, operators and verifiers shall comply with the following criteria:

a. Projects 'under construction' and 'as built' shall be labelled 'awaiting verification' until they have been verified by an independent third party, after which they shall be labelled 'verified'.
b. Projects 'awaiting verification' shall be rendered 'void' after five years unless they have been verified by an independent third party following the criteria in this paragraph (3.3).
c. Operators of projects that are rendered 'void' due to lack of verification lose any rights connected to certification following this Certification Protocol. These rights shall be re-instated once the projects have been labelled 'verified'.
d. For projects 'under construction' to be labelled as 'verified', data, proofs and descriptions submitted by operators according to the criteria outlined in paragraph 3.1b shall be verified by an independent third party with relevant accreditation and/or expertise on the subject of auditing or verification of LCAs, building procedures or building environmental performance (MPG).
e. Projects 'under construction' that have been verified shall be re-verified once they are labelled as 'as built'.
f. For projects 'as built' to be labelled as 'verified', data, proofs and descriptions submitted by operators according to the criteria outlined in paragraph 3.1c shall be verified by an independent third party with relevant accreditation and/or expertise on the subject of auditing or verification of LCAs, building procedures or building environmental performance (MPG).
g. When deemed necessary by the verification body, certification body or certification scheme, operators of projects 'as built' may be requested to submit an updated MPG based on the final realisation of the construction project.